2SK715

N-Channel JFET
15V, 7.3 to 24mA, 50mS

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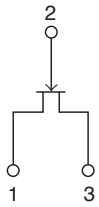
Applications

- AM Tuner RF Amp, Low-noise Amp
- HF Low-noise Amp

Features

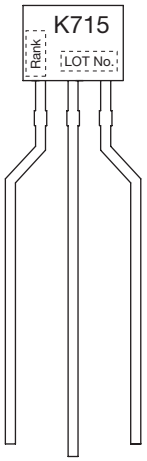
- Adoption of FBET Process
- Large $|yfs|$
- Small Ciss
- Very Low Noise Figure

Electrical Connection



1 : Source
2 : Gate
3 : Drain

Marking



2SK715U-AC
2SK715V-AC
2SK715W-AC

TO-92-3 / SPA-WA

Specifications

Absolute Maximum Ratings at Ta=25°C

Parameter	Symbol	Ratings	Unit
Drain-to-Source Voltage	V_{DSS}	15	V
Gate-to-Drain Voltage	V_{GDS}	−15	V
Gate Current	I_G	10	mA
Drain Current	I_D	50	mA
Allowable Power Dissipation	P_D	300	mW
Junction Temperature	Tj	125	°C
Storage Temperature	Tstg	−55 to +125	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

Electrical Characteristics at Ta=25°C

Parameter	Symbol	Conditions	Ratings			Unit
			min	typ	max	
Gate-to-Drain Breakdown Voltage	$V_{(BR)GDS}$	I_G=−10μA, V_{DS}=0V	−15			V
Gate-to-Source Leakage Current	I_{GSS}	V_{GS}=−10V, V_{DS}=0V			−1.0	nA
Zero-Gate Voltage Drain Current	I_{DSS}*	V_{DS}=5V, V_{GS}=0V	7.3*		24.0*	mA
Cutoff Voltage	$V_{GS(off)}$	V_{DS}=5V, I_D=100μA		−0.6	−1.4	V
Forward Transfer Admittance	$\|yfs\|$	V_{DS}=5V, V_{GS}=0V, f=1kHz	25	50		mS
Input Capacitance	Ciss	V_{DS}=5V, V_{GS}=0V, f=1MHz		10		pF
Reverse Transfer Capacitance	Crss			3.0		pF
Noise Figure	NF	V_{DS}=5V, R_g=1kΩ, I_D=1mA, f=1kHz		1.5		dB

* : The 2SK715 is classified by I_{DSS} as follows : (unit : mA)

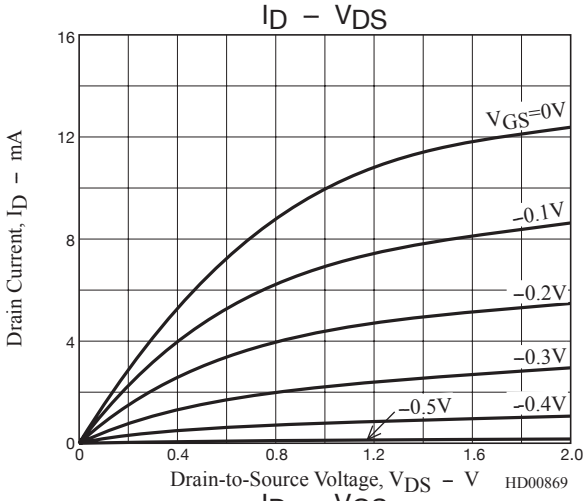
Rank	U	V	W
I_{DSS}	7.3 to 12.0	10.0 to 17.0	14.5 to 24.0

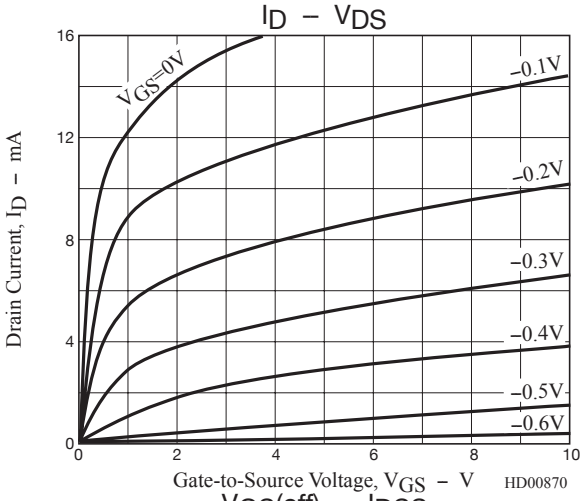
Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

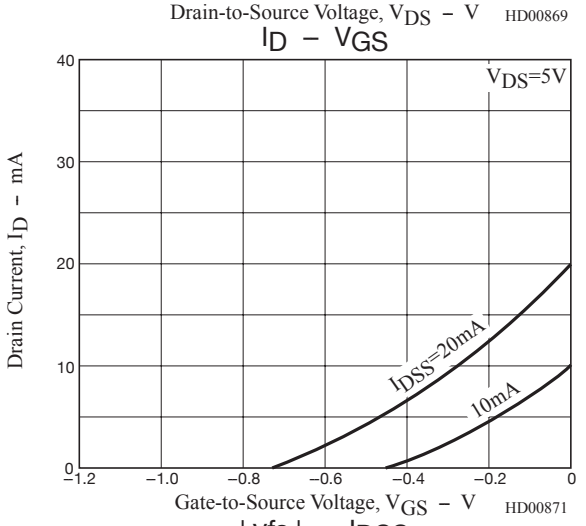
ORDERING INFORMATION

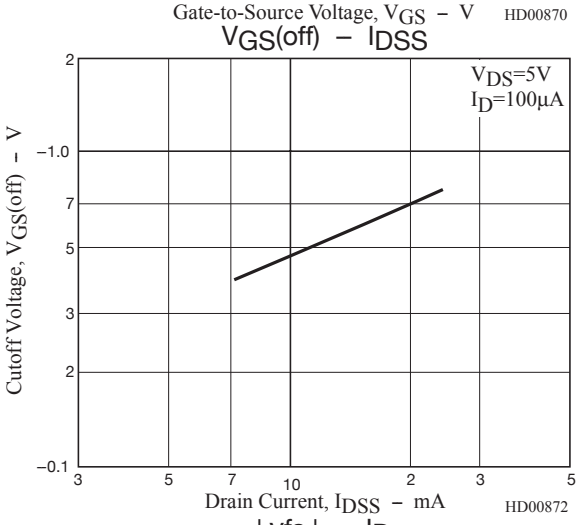
See detailed ordering and shipping information on page 5 of this data sheet.

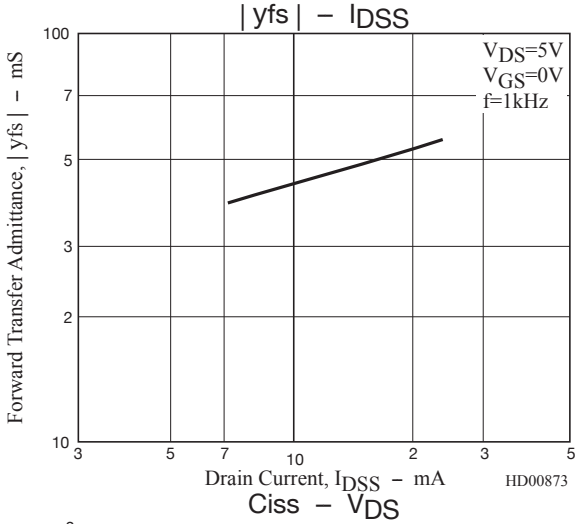
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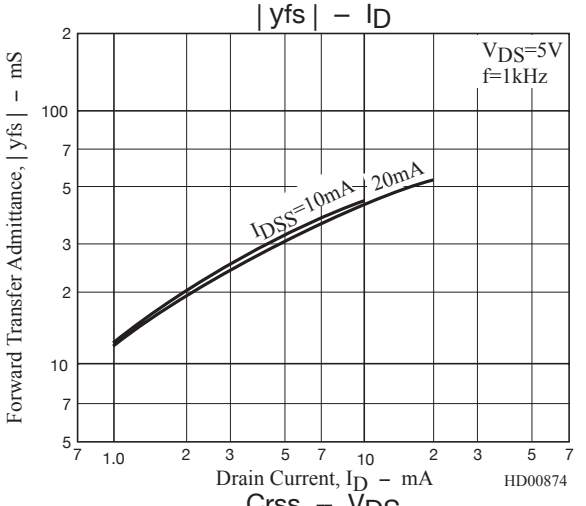

I_D – V_{DS}
Drain Current, I_D – mA
Drain-to-Source Voltage, V_{DS} – V
V_{GS}=0V
–0.1V
–0.2V
–0.3V
–0.4V
–0.5V
HD00869

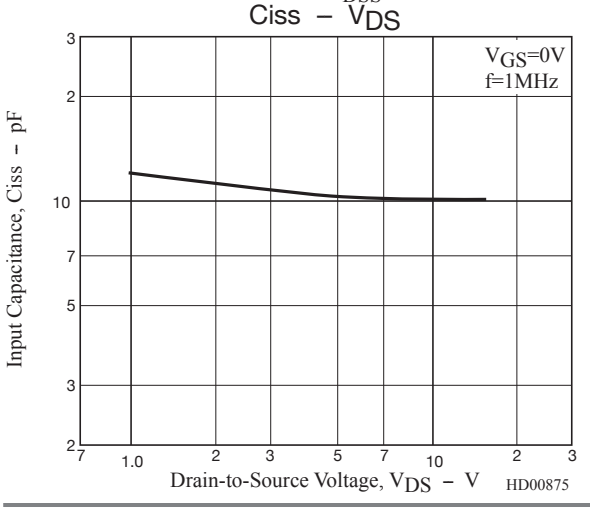

I_D – V_{DS}
Drain Current, I_D – mA
Gate-to-Source Voltage, V_{GS} – V
V_{GS}=0V
–0.1V
–0.2V
–0.3V
–0.4V
–0.5V
–0.6V
HD00870

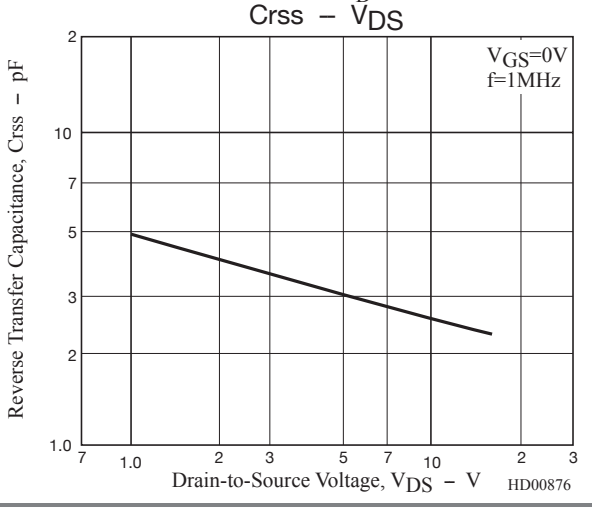

I_D – V_{GS}
Drain Current, I_D – mA
Gate-to-Source Voltage, V_{GS} – V
V_{DS}=5V
I_{DSS}=20mA
10mA
HD00871


$V_{GS(off)}$ – I_{DSS}
Cutoff Voltage, $V_{GS(off)}$ – V
Drain Current, I_{DSS} – mA
V_{DS}=5V
I_D=100µA
HD00872

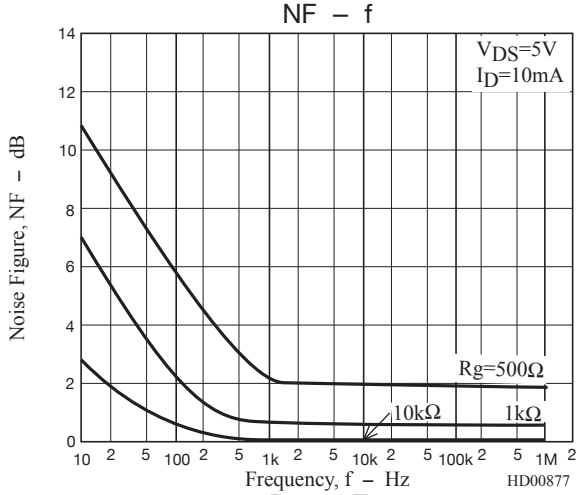

| yfs | – I_{DSS}
Forward Transfer Admittance, | yfs | – mS
Drain Current, I_{DSS} – mA
V_{DS}=5V
V_{GS}=0V
f=1kHz
HD00873


| yfs | – I_D
Forward Transfer Admittance, | yfs | – mS
Drain Current, I_D – mA
V_{DS}=5V
f=1kHz
I_{DSS}=10mA
20mA
HD00874

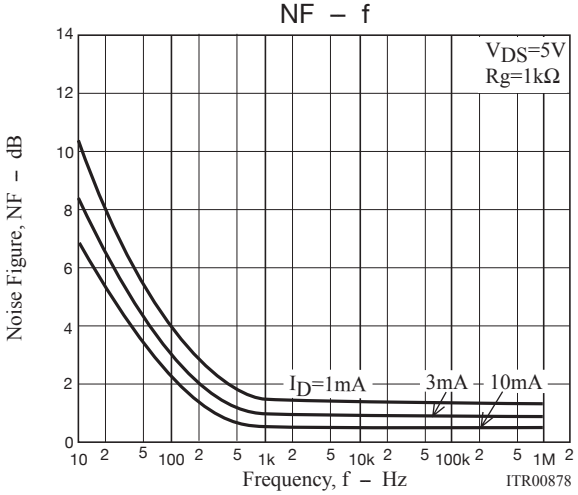

Ciss – V_{DS}
Input Capacitance, Ciss – pF
Drain-to-Source Voltage, V_{DS} – V
V_{GS}=0V
f=1MHz
HD00875


Crss – V_{DS}
Reverse Transfer Capacitance, Crss – pF
Drain-to-Source Voltage, V_{DS} – V
V_{GS}=0V
f=1MHz
HD00876

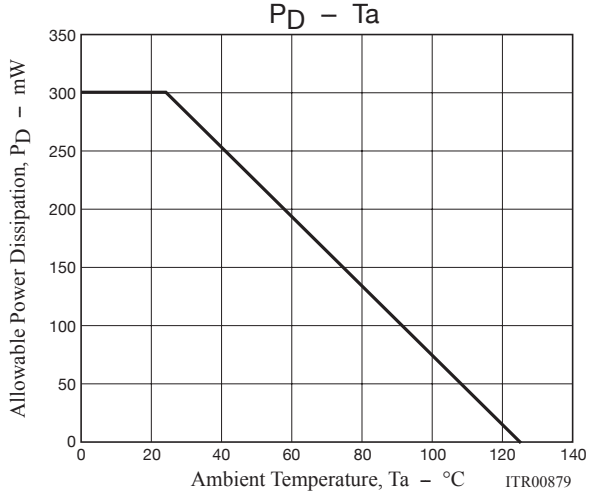
NF – f


VDS=5V
ID=10mA
Rg=500Ω
10kΩ
1kΩ
Noise Figure, NF – dB
Frequency, f – Hz
HD00877

NF – f


VDS=5V
Rg=1kΩ
ID=1mA
3mA
10mA
Noise Figure, NF – dB
Frequency, f – Hz
ITR00878

PD – Ta

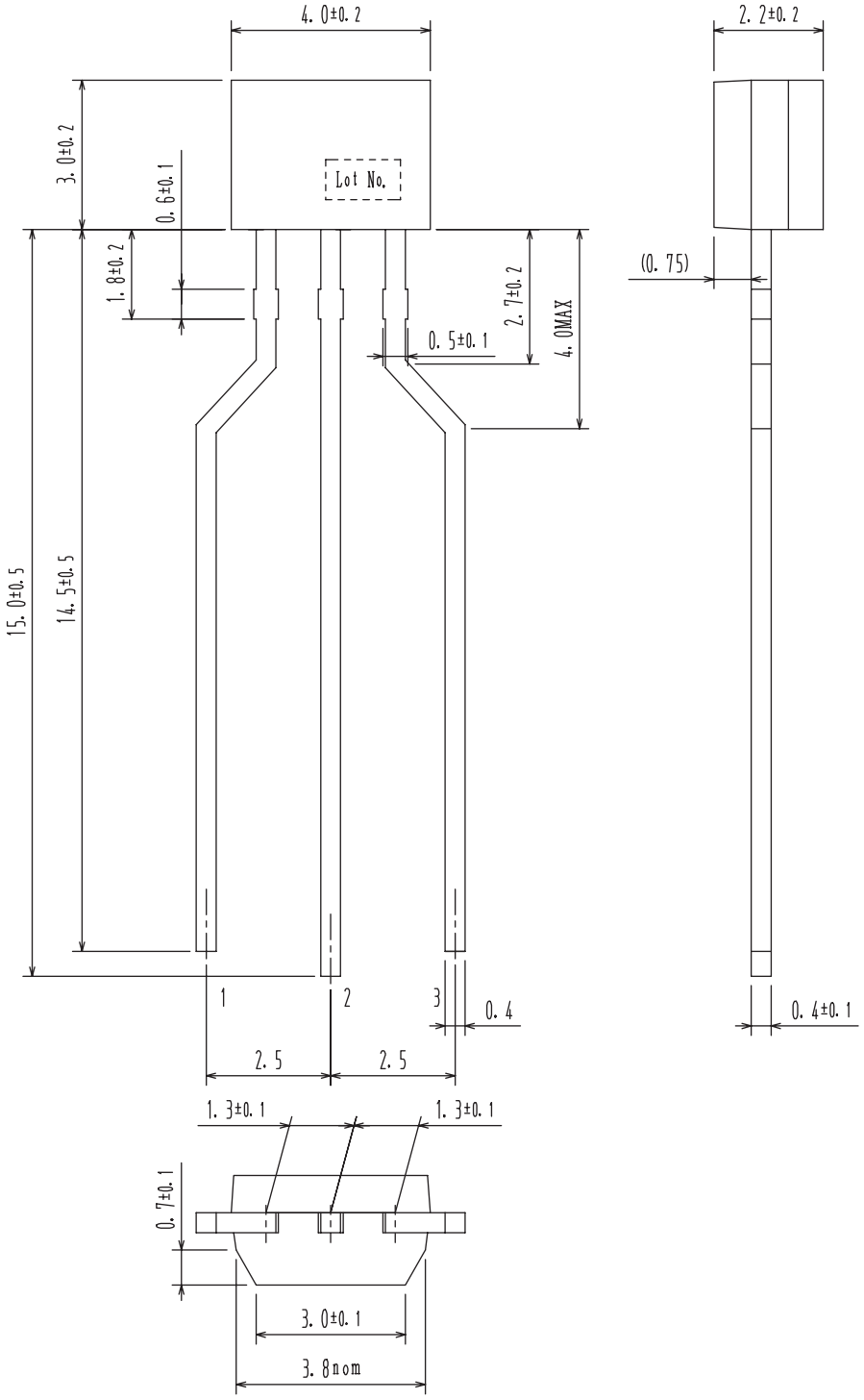

Allowable Power Dissipation, PD – mW
Ambient Temperature, Ta – °C
ITR00879

Package Dimensions

unit : mm

2SK715U-AC, 2SK715V-AC, 2SK715W-AC

TO-92 3 3.0x4.0 / SPA-WA
CASE 135AK
ISSUE O



ORDERING INFORMATION

Device	Package	Shipping	memo
2SK715U-AC	TO-92-3 / SPA-W	2,500pcs. / Tape and Reel	Pb-Free
2SK715V-AC			
2SK715W-AC			

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